

Data Cleaning and Aggregation Rules

Study: Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision

This document reproduces the full set of inclusion and exclusion criteria applied during data cleaning and the rules by which response categories were recoded and aggregated. It is derived from the internal cleaning log (Versions 1.0 and 1.1). Column letters (e.g. Column J) refer to the raw Microsoft Forms export. Respondent identifiers are the row numbers of the raw export. Free-text responses are reproduced verbatim in German, since they are the data on which the coding decisions were made; assigned categories are stated in English following the questionnaire translation.

1. Sample derivation overview

Cleaning proceeded in two stages. Stage 1 (Version 1.0 of the log) removed records that did not meet the inclusion criteria, yielding the cleaned data set (dataAfterCleaning.csv, n = 363). Stage 2 (Version 1.1) prepared the analytical sample by excluding records without an answer to the core Shadow AI item, aggregating and recoding response categories, and excluding heterogeneous low-frequency work settings.

Stage	Excluded (n)	Remaining (N)
Raw export		403
Duplicate check (no duplicates found)	0	403
Stage 1: federal state (Column J)	7	396
Stage 1: professional group (Column M)	10	386
Stage 1: work setting (Column L)	19	367
Stage 1: single-field exclusions (age, gender, guidelines, reasons for non-use)	4	363
Cleaned data set (dataAfterCleaning.csv)		363
Stage 2: no answer to core Shadow AI item (Column BD)	28	335
Stage 2: aggregated low-frequency work settings ('Other')	15	320
Final analytical sample		320

2. Stage 1: validity exclusions (Version 1.0)

Records were excluded where the respondent did not belong to the target population (physicians practising in Germany), was no longer professionally active, or had not provided answers on key variables. Naming adjustments did not change the sample size.

2.1 Duplicate check

The data set was checked for duplicate submissions. None were present.

2.2 Federal state (Column J): 7 excluded

Excluded because the place of work was outside Germany, the respondent was retired, or no federal state was given:

- Austria: record 163
- Switzerland: records 259, 370, 402
- Retired: record 371

- No answer: records 112, 168

Remaining: n = 396.

2.3 Professional group (Column M): 10 excluded, 2 adjusted

Excluded because the respondent was not a physician:

- Other healthcare professionals: records 14, 96, 132, 174
- Professional nursing: records 130, 136, 142
- Retired person: record 349
- Medical assistant (MFA): record 363
- Medical student: record 367

Naming adjustments (no exclusion): the free-text entry of record 1 was assigned to 'Physician'; record 256 was recoded from 'Arzt' to 'Physician'.

Remaining: n = 386.

2.4 Work setting (Column L): 19 excluded

Excluded because the setting was not clinical practice or the respondent was no longer practising:

- Self-governance / politics / professional association: records 24, 27, 65, 165, 256
- Administration / management: records 190, 229, 243, 268
- Further exclusions for combinations of missing data and indications of no longer practising (10 records): record 12 (lecturer); record 35 (works as a freelance translator); records 68, 149, 155 (excluded on the basis of age and extensive missingness across many fields, indicating they are presumably no longer working, with 'too old' given as free text and age over 70); record 189 (retired, Column AW); records 196, 249 (retired, Column AP); record 271 (retired, Column BB); record 384 (Column AN)

Remaining: n = 367.

2.5 Single-field exclusions and one adjustment

- Age (Column I): record 94, no answer. Excluded.
- Gender (Column K): record 158, no answer. Excluded.
- Guidelines (Column BE): record 251, not professionally active. Excluded.
- Reasons for not using AI (Column AN): record 241, retired. Excluded.
- Training (Column AT): record 62 adjusted, not excluded. The respondent gave no answer and did not complete the dependent follow-up fields (Columns AU and AV); the entry was set to 'No'.

Remaining after Stage 1: n = 363 (dataAfterCleaning.csv).

Variables retained without exclusion at this stage: specialty (Column N) retained one 'no answer' record; the Shadow AI item (Column BD) retained its 'no answer' records at this stage (these were addressed in Stage 2). Provision of AI (Column AL), AI-use frequency (Column AQ) and application areas (Column AR) were not used for exclusion.

3. Stage 2: analytical exclusions, recoding and aggregation (Version 1.1)

3.1 Shadow AI item (Column BD)

For the main analyses, the 28 records without an answer to the core Shadow AI item were excluded, reducing the sample from 363 to 335.

3.2 Work setting aggregation (Column L)

Conceptually related categories were aggregated:

- Hospital (inpatient) and Hospital (outpatient) were combined into 'Hospital'.
- Practice (group practice / joint medical practice) and Practice (single practice) were combined into 'Practice'.
- Public health service, emergency medical service and 'no answer' were combined into 'Other' (n = 15).
- Medical care centre (MVZ) was retained as a separate category.

The 'Other' group (n = 15) was excluded from the final analytical sample owing to insufficient cell counts for inferential testing, reducing the sample from 335 to 320. The three settings carried into analysis are Hospital, Practice and Medical care centre.

3.3 Guidelines recoding (Column BE)

Free-text responses were recoded to the questionnaire categories Yes, No or Don't know. Respondents who indicated that they are themselves the employer or are self-employed were coded according to whether a guideline existed (Yes where one had been issued, No where none existed). Non-substantive placeholder entries ('0', '@', and a bare reference to a data protection regulation) were coded as Don't know.

ID	Assigned	Original response (verbatim)
1	Don't know	0
74	Don't know	0
76	Yes	ich bin der Arbeitgeber
78	No	wird eben erarbeitet
91	Don't know	@
101	No	Keine Anwendung, Selbständig
111	Yes	selbständig
143	Don't know	Datenschutzverordnung
154	No	Haben wir nicht, daher noch nicht implementiert.
201	No	Keine Nutzungsfreigabe
202	Yes	bin selbst Praxisinhaber
216	No	Verboten.
246	No	KI-Schulung für alle Mitarbeiterinnen geplant
258	No	nein, obige fragen sind teils imperativ gestellt bei den Auswahlmöglichkeiten der bisher genutzten KI Möglichkeiten. Es waren keine Möglichkeiten, anzugeben, ob man überhaupt das System benutzen möchte.
275	No	eigene Praxis
313	Yes	Habe selbst eine initiiert zum Umgang mit KI bei Dissertationen
337	Yes	Bin mein eigener Arbeitgeber
347	No	Selber Arbeitgeber
348	Yes	bin mein eigener Arbeitgeber
372	Don't know	0

3.4 Reasons for not using AI (Column AN)

A new category, 'No concerns / already uses AI', was added for respondents whose free text indicated that they do in fact use AI, so that the original item (reasons for non-use) did not apply to them. Other free-text responses were assigned to the existing questionnaire categories. Where a respondent selected several categories, all are listed, separated by a semicolon.

ID	Assigned category	Original response (verbatim)
10	Not available	Kosten und fehlende Software
11	Not available	Noch nicht etabliert
55	Lack of knowledge / training; No perceived usefulness / time saving	Vielleicht bin ich mir dessen Einsatz nicht im Klaren
63	No concerns / already uses AI	Keine passende Antwort
70	Not available	Bin angestellt, nicht meine Entscheidung
76	Security / data protection concerns; No perceived usefulness / time saving	in manchen Bereichen ist der menschliche Kontakt besser. Wir nutzen KI dort wo es das Leben leichter und die Behandlung sicherer macht.
84	No concerns / already uses AI	Frage fehlerhaft, ich nutze sie doch siehe oben
85	No concerns / already uses AI	Ich nutze sie nur nicht vom Arbeitgeber sondern über private Anmeldung und finanzierung
100	No concerns / already uses AI	Keine
109	No concerns / already uses AI	Ich nutze KI-basierte Anwendungen - siehe oben!
113	No concerns / already uses AI	die frage versteh ich nicht, da ich KI nutze
119	No concerns / already uses AI	-
120	No concerns / already uses AI	ich setze es ein
131	No concerns / already uses AI	Nutzen sie
143	No concerns / already uses AI	Ich benutze doch welche
153	No concerns / already uses AI	Wir nutzen KI
154	Not available; Lack of capacity / time for implementation	z.T. treffen alle o.g. Punkte zu
166	Not available; No perceived usefulness / time saving; Lack of knowledge / training	viel Zeitaufwand für Einarbeitung
202	No concerns / already uses AI	nutze KI
204	No concerns / already uses AI	Nicht zu beantworten, da ich KI nutze
222	No concerns / already uses AI	Nicht relevant
234	Not available; Security / data protection concerns; Lack of capacity / time for implementation	fehlendes Geld
242	Lack of capacity / time for implementation	Kosten
258	Lack of integration options in existing systems; No perceived usefulness / time saving	kein bedarf

ID	Assigned category	Original response (verbatim)
265	No concerns / already uses AI	Ich nutze sie.
304	No concerns / already uses AI	Testphase
307	No concerns / already uses AI	ich nutze KI
310	Not available	Als sporadisch in der Notfallpraxis tätiger Arzt bisher nicht erforderlich für mich, wird auch nicht bereitgestellt
312	Lack of integration options in existing systems; Security / data protection concerns	Die Firmen, die Programme für die Blutbanklogistik entwickeln, haben KI noch nicht in deren vorhandenen Programme implementiert.
319	Not available; Lack of integration options in existing systems	häufig bringen die Pat. z.B. keine Medikamentenpläne mit QR-Code mit sondern lieber handgeschriebene Zettelchen
321	No concerns / already uses AI	Ich nutze KI-Tools!!!
322	Not available; Lack of capacity / time for implementation	Kh hat kein Geld
324	No concerns / already uses AI	Nutze ich ja
327	Lack of capacity / time for implementation	Kosten
354	No perceived usefulness / time saving	Schon das PVS System ist so mangelhaft, das heißt schon die Basis funktioniert nicht, ich benutze meine privaten Accounts
379	No perceived usefulness / time saving	Das Fehlerrisiko ist mir noch zu hoch
398	No perceived usefulness / time saving	zur Zeit wenig Evidenz

3.5 Application areas (Column AR)

Free-text 'Other' responses to the multiple-choice item on application areas were assigned to the five standard categories (Diagnostics and findings; Treatment planning and therapy decisions; Administrative tasks; Research; Patient communication and service) where the text denoted such an area, or to 'No answer' where it did not. The table lists the assigned categories and the distinctive free text on which the assignment was based.

ID	Assigned category	Original free text (verbatim)
10	Diagnostics and findings	zusätzliche Auswertung der Lunge
42	Administrative tasks	Exposition
45	Research	RECHERCHE
83	Administrative tasks	Erstellen von Prüfungsfragen
100	Treatment planning and therapy decisions; Administrative tasks	Bei der Strahlentherapie selbst
102	No answer	Nur wenig Nützliches bei uns vorhanden, für Dienstplan etc. kein Geld zur Verfügung gestellt
121	Diagnostics and findings; Treatment planning and therapy decisions	Unterstützung bei seltenen Diagnosen zur Unterstützung bei Diagnose- und Therapiefindung
129	Administrative tasks	Dokumentation / Spracherkennung

ID	Assigned category	Original free text (verbatim)
295	Administrative tasks	Diktiergerät
319	Administrative tasks; Diagnostics and findings	chatGPT für DRG-Codes, Amboss zum Nachlesen, chatGPT zum Recherchieren mancher Sachen, Google Translate zum Übersetzen
331	Diagnostics and findings	OpenEvidence
335	Administrative tasks	Behandlungsdokumentation (duplicate selection collapsed to a single category)
336	No answer	Große Unsicherheit, wie sicher die Anwendungen sind. Ich kenne keine speziellen und akzeptierten KI-Anwendungen in der Medizin
337	Administrative tasks; Treatment planning and therapy decisions; Research; Diagnostics and findings	Spracherkennung und Literaturrecherche
339	Diagnostics and findings	nur mal so zum Nachschlagen und bestätigen meiner vorhandenen Überlegungen
342	Treatment planning and therapy decisions; Patient communication and service; Administrative tasks	Übersetzungen für fremdsprachliche Patienten
354	Research; Diagnostics and findings; Treatment planning and therapy decisions	Ich benutze meine privaten Account

4. Final analytical sample and analysis code

After Stage 2, the final analytical sample comprised 320 records. Descriptive, governance-index and regression analyses were conducted in Python (version 3.12.13) using pandas, scipy.stats and statsmodels, executed in Google Colab. The analysis notebook should be deposited in the repository in anonymised form (see the editor's note on the first page).